

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7				- Pepsin in stomach, hydrolyses peptide bonds breaking down polypeptides into shorter chains of amino acids. - The pancreas produces proteases such as trypsin which breaks down polypeptide chains into shorter chains. - Cells in small intestine secrete peptidases which complete the breakdown of polypeptides into amino acids. - Ref to exopeptidases and endopeptidases. - Amino acids absorbed into the blood from small intestine transported to the muscles. - Cows large numbers of bacteria are produced in the first three chambers of the 'stomach' - make protein using urea - when the bacteria pass into the true stomach they are killed by the acid. - Proteins {in/ from} the bacteria are then digested and absorbed - Horses do not have saliva containing urea - This explains why horses need more protein in food than cows - In horses the bacteria are found in the caecum/ large intestine - Protein in these bacteria are lost in the faeces because no digestion or absorption takes place in the large intestine. - This explains why horse manure has a higher levels of organic nitrogen.	3	4	2	9		
				7-9 marks Detailed explanation of protein digestion Explanation of use of urea by bacteria in cow/ digestion of bacteria Explanation of increased protein in diet of horse/ nitrogen content in manure of horse <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i>						

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					AO1	AO2	AO3	Total	Maths	Prac
				4-6 marks Any two from: Explanation of protein digestion Brief explanation of use of urea by bacteria in cow/ digestion of bacteria Brief explanation of increased protein in diet/ nitrogen content in manure <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i>						
				1-3 marks. Any one from: Brief explanation of protein digestion Brief explanation of ruminant digestion Brief explanation of increased protein in diet/ nitrogen content in manure <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i>						
				0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 7 total	3	4	2	9	0	0